

DBMS Lab Programs

Program 1: Consider the schema for College Database

STUDENT(USN, SName, Address, Phone, Gender)

SEMSEC(SSID, Sem, Sec) CLASS(USN, SSID)

COURSE(Subcode, Title, Sem, Credits)

IAMARKS(USN, Subcode, SSID, Test1, Test2, Test3, FinalIA)

Write SQL queries to :

1. List all the student details studying in fourth semester 'C' section.

2. Compute the total number of male and female students in each semester and in each section.

3. Create a view of Test1 marks of student USN '4MH22CS200' in all Courses.

4. Calculate the FinalIA (average of best two test marks) and update the corresponding table for all students.

5. Categorize students based on the following criterion: If FinalIA = 17 to 20 then CAT = 'Outstanding' If FinalIA = 12 to 16 then CAT = 'Average' If FinalIA < 12 then CAT = 'Weak' Give these details only for 4th semester ALL section students.

```
CREATE TABLE STUDENT (  
    USN VARCHAR(20) PRIMARY KEY,  
    SName VARCHAR(100),  
    Address VARCHAR(255),  
    Phone VARCHAR(15),  
    Gender VARCHAR(10)  
);
```

```
mysql> DESC STUDENT;  
+-----+-----+-----+-----+-----+-----+  
| Field | Type          | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| USN   | varchar(20)   | NO   | PRI | NULL    |       |  
| SName | varchar(100)  | YES  |     | NULL    |       |  
| Address | varchar(255) | YES  |     | NULL    |       |  
| Phone | varchar(15)   | YES  |     | NULL    |       |  
| Gender | varchar(10)  | YES  |     | NULL    |       |  
+-----+-----+-----+-----+-----+-----+  
5 rows in set (0.19 sec)
```

```
CREATE TABLE SEMSEC (
    SSID INT PRIMARY KEY,
    Sem INT,
    Sec CHAR(1)
);
```

```
mysql> DESC SEMSEC;
+-----+-----+-----+-----+-----+-----+
| Field | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| SSID  | int       | NO   | PRI | NULL    |       |
| Sem   | int       | YES  |     | NULL    |       |
| Sec   | char(1)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
CREATE TABLE CLASS (
    USN VARCHAR(20),
    SSID INT,
    PRIMARY KEY (USN, SSID),
    FOREIGN KEY (USN) REFERENCES STUDENT(USN),
    FOREIGN KEY (SSID) REFERENCES SEMSEC(SSID)
);
```

```
mysql> DESC CLASS;
+-----+-----+-----+-----+-----+-----+
| Field | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| USN   | varchar(20) | NO   | PRI | NULL    |       |
| SSID  | int         | NO   | PRI | NULL    |       |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

```
CREATE TABLE COURSE (
    Subcode VARCHAR(10) PRIMARY KEY,
```

Title VARCHAR(100),
Sem INT,
Credits INT
);

```
mysql> DESC COURSE;
```

Field	Type	Null	Key	Default	Extra
Subcode	varchar(10)	NO	PRI	NULL	
Title	varchar(100)	YES		NULL	
Sem	int	YES		NULL	
Credits	int	YES		NULL	

4 rows in set (0.01 sec)

CREATE TABLE IAMARKS (
USN VARCHAR(20),
Subcode VARCHAR(10),
SSID INT,
Test1 INT,
Test2 INT,
Test3 INT,
FinalIA INT,
PRIMARY KEY (USN, Subcode, SSID),
FOREIGN KEY (USN) REFERENCES STUDENT(USN),
FOREIGN KEY (Subcode) REFERENCES COURSE(Subcode),
FOREIGN KEY (SSID) REFERENCES SEMSEC(SSID)
);

```
mysql> DESC IAMARKS;
```

Field	Type	Null	Key	Default	Extra
USN	varchar(20)	NO	PRI	NULL	
Subcode	varchar(10)	NO	PRI	NULL	
SSID	int	NO	PRI	NULL	
Test1	int	YES		NULL	
Test2	int	YES		NULL	
Test3	int	YES		NULL	
FinalIA	int	YES		NULL	

```
7 rows in set (0.01 sec)
```

INSERT INTO STUDENT (USN, SName, Address, Phone, Gender)

VALUES

('4MH22CS200', 'John Doe', '123 ABC Street, City', '9876543210', 'Male'),

('4MH22CS201', 'Jane Smith', '456 XYZ Avenue, City', '9876543211', 'Female'),

('4MH22CS202', 'Alice Brown', '789 DEF Road, City', '9876543212', 'Female'),

('4MH22CS203', 'Bob White', '101 GHI Lane, City', '9876543213', 'Male');

```
mysql> SELECT *FROM STUDENT;
```

USN	SName	Address	Phone	Gender
4MH22CS200	John Doe	123 ABC Street, City	9876543210	Male
4MH22CS201	Jane Smith	456 XYZ Avenue, City	9876543211	Female
4MH22CS202	Alice Brown	789 DEF Road, City	9876543212	Female
4MH22CS203	Bob White	101 GHI Lane, City	9876543213	Male

```
4 rows in set (0.01 sec)
```

INSERT INTO SEMSEC (SSID, Sem, Sec)

VALUES

(1, 1, 'A'), -- First semester, Section A

(2, 1, 'B'), -- First semester, Section B

(3, 4, 'A'), -- Fourth semester, Section A

(4, 4, 'C'); -- Fourth semester, Section C

```
mysql> SELECT *FROM SEMSEC;
+-----+-----+-----+
| SSID | Sem | Sec |
+-----+-----+-----+
| 1 | 1 | A |
| 2 | 1 | B |
| 3 | 4 | A |
| 4 | 4 | C |
+-----+-----+-----+
4 rows in set (0.02 sec)
```

INSERT INTO CLASS (USN, SSID)

VALUES

('4MH22CS200', 1), -- John Doe in First semester, Section A

('4MH22CS201', 1), -- Jane Smith in First semester, Section A

('4MH22CS202', 3), -- Alice Brown in Fourth semester, Section A

('4MH22CS203', 4); -- Bob White in Fourth semester, Section C

```
mysql> SELECT *FROM CLASS;
+-----+-----+
| USN | SSID |
+-----+-----+
| 4MH22CS200 | 1 |
| 4MH22CS201 | 1 |
| 4MH22CS202 | 3 |
| 4MH22CS203 | 4 |
+-----+-----+
4 rows in set (0.01 sec)
```

INSERT INTO COURSE (Subcode, Title, Sem, Credits)

VALUES

('CS101', 'Introduction to Programming', 1, 4), -- First semester course

('CS102', 'Data Structures', 1, 3), -- First semester course

('CS201', 'Operating Systems', 4, 4), -- Fourth semester course

('CS202', 'Database Management', 4, 3); -- Fourth semester course

```
mysql> SELECT *FROM COURSE;
```

Subcode	Title	Sem	Credits
CS101	Introduction to Programming	1	4
CS102	Data Structures	1	3
CS201	Operating Systems	4	4
CS202	Database Management	4	3

4 rows in set (0.01 sec)

INSERT INTO IAMARKS (USN, Subcode, SSID, Test1, Test2, Test3, FinalIA)

VALUES

('4MH22CS200', 'CS101', 1, 18, 15, 17, 17), -- John Doe's marks for CS101 in Semester 1

('4MH22CS200', 'CS102', 1, 16, 18, 14, 16), -- John Doe's marks for CS102 in Semester 1

('4MH22CS201', 'CS101', 1, 19, 17, 16, 17), -- Jane Smith's marks for CS101 in Semester 1

('4MH22CS201', 'CS102', 1, 14, 13, 18, 15), -- Jane Smith's marks for CS102 in Semester 1

('4MH22CS202', 'CS201', 3, 20, 18, 17, 18), -- Alice Brown's marks for CS201 in Semester 4

('4MH22CS202', 'CS202', 3, 15, 18, 20, 18), -- Alice Brown's marks for CS202 in Semester 4

('4MH22CS203', 'CS201', 4, 12, 14, 10, 12), -- Bob White's marks for CS201 in Semester 4

('4MH22CS203', 'CS202', 4, 13, 15, 12, 14); -- Bob White's marks for CS202 in Semester 4

```
mysql> SELECT *FROM IAMARKS;
```

USN	Subcode	SSID	Test1	Test2	Test3
4MH22CS200	CS101	1	18	15	17
4MH22CS200	CS102	1	16	18	14
4MH22CS201	CS101	1	19	17	16
4MH22CS201	CS102	1	14	13	18
4MH22CS202	CS201	3	20	18	17
4MH22CS202	CS202	3	15	18	20
4MH22CS203	CS201	4	12	14	10
4MH22CS203	CS202	4	13	15	12

```
8 rows in set (0.01 sec)
```

1. List all the student details studying in fourth semester 'C' section.

```
SELECT S.USN, S.SName, S.Address, S.Phone, S.Gender
FROM STUDENT S,CLASS C,SEMSEC SS
WHERE S.USN = C.USN
AND C.SSID = SS.SSID
AND SS.Sem = 4 AND SS.Sec = 'C';
```

```
mysql> SELECT S.USN, S.SName, S.Address, S.Phone, S.Gender
FROM STUDENT S,CLASS C,SEMSEC SS
WHERE S.USN = C.USN
AND C.SSID = SS.SSID
AND SS.Sem = 4 AND SS.Sec = 'C';
```

USN	SName	Address	Phone	Gender
4MH22CS203	Bob White	101 GHI Lane, City	9876543213	Male

```
1 row in set (0.01 sec)
```

2. Compute the total number of male and female students in each semester and in each section.

```
SELECT SS.SEM, SS.SEC, S.GENDER, COUNT(*) AS COUNT
FROM STUDENT S,CLASS C ,SEMSEC SS
WHERE S.USN = C.USN
AND SS.SSID = C.SSID
GROUP BY SS.SEM, SS.SEC, S.GENDER
```

ORDER BY SS.SEM;

SEM	SEC	GENDER	COUNT
1	A	Female	1
1	A	Male	1
4	A	Female	1
4	C	Male	1

4 rows in set (0.03 sec)

3.Create a view of Test1 marks of student USN '1BI15CS101' in all Courses.

```
CREATE VIEW Test_Marks AS
SELECT IAM.USN, IAM.Subcode, IAM.Test1
FROM IAMARKS IAM
WHERE IAM.USN = '4MH22CS200';
```

```
mysql> select *from TEST_MARKS;
```

USN	Subcode	Test1
4MH22CS200	CS101	18
4MH22CS200	CS102	16

2 rows in set (0.01 sec)

4. Calculate the FinalIA (average of best two test marks) and update the corresponding table for all students.

```
UPDATE IAMARKS SET
```

```
FINALIA=GREATEST(TEST1+TEST2,TEST2+TEST3,TEST1+TEST3)/2;
```

```
mysql> SELECT *FROM IAMARKS;
```

USN	Subcode	SSID	Test1	Test2	Test3	FinalIA
4MH22CS200	CS101	1	18	15	17	18
4MH22CS200	CS102	1	16	18	14	17
4MH22CS201	CS101	1	19	17	16	18
4MH22CS201	CS102	1	14	13	18	16
4MH22CS202	CS201	3	20	18	17	19
4MH22CS202	CS202	3	15	18	20	19
4MH22CS203	CS201	4	12	14	10	13
4MH22CS203	CS202	4	13	15	12	14

```
8 rows in set (0.00 sec)
```

5. Categorize students based on the following criterion: If FinalIA = 17 to 20 then CAT = 'Outstanding' If FinalIA = 12 to 16 then CAT = 'Average' If FinalIA < 12 then CAT = 'Weak' Give these details only for 4th semester ALL section students

```
SELECT S.USN, S.SName, IAM.FinalIA,
CASE
    WHEN IAM.FinalIA BETWEEN 17 AND 20 THEN 'Outstanding'
    WHEN IAM.FinalIA BETWEEN 12 AND 16 THEN 'Average'
    WHEN IAM.FinalIA < 12 THEN 'Weak'
END AS Category
FROM STUDENT S, CLASS C, SEMSEC SS, IAMARKS IAM
WHERE S.USN = C.USN
AND C.SSID = SS.SSID
AND S.USN = IAM.USN AND SS.SSID = IAM.SSID
AND SS.Sem = 4;
```

```
mysql> SELECT S.USN, S.SName, IAM.FinalIA, Category
```

USN	SName	FinalIA	Category
4MH22CS202	Alice Brown	19	Outstanding
4MH22CS202	Alice Brown	19	Outstanding
4MH22CS203	Bob White	13	Average
4MH22CS203	Bob White	14	Average

```
4 rows in set (0.01 sec)
```

2	Consider the schema for Movie Database: ACTOR(<i>Act_id, Act_Name, Act_Gender</i>) DIRECTOR(<i>Dir_id, Dir_Name, Dir_Phone</i>) MOVIES(<i>Mov_id, Mov_Title, Mov_Year, Mov_Lang, Dir_id</i>) MOVIE_CAST(<i>Act_id, Mov_id, Role</i>) RATING(<i>Mov_id, Rev_Stars</i>) Write SQL queries to 1. List the titles of all movies directed by 'Mani Rathnam'. 2. Find the movie names where one or more actors acted in two or more movies. 3. List all actors who acted in a movie before 2000 and also in a movie after 2024(use JOIN operation). 4. Find the title of movies and number of stars for each movie that has at least one rating and find the highest number of stars that movie received. Sort the result by movie title. 5. Update rating of all movies directed by 'Shankar' to 5.
---	--

```
CREATE TABLE ACTOR (
```

```
Act_id INT PRIMARY KEY,
```

```
Act_Name VARCHAR (25) NOT NULL,
```

```
Act_Gender CHAR (10) NOT NULL
```

```
);
```

```
CREATE TABLE DIRECTOR (
```

```
Dir_id INT PRIMARY KEY,
```

```
Dir_Name VARCHAR(25) NOT NULL,
```

```
Dir_Phone CHAR(10) NOT NULL
```

```
);
```

```
CREATE TABLE MOVIES (
```

```
Mov_id INT PRIMARY KEY
```

```
Mov_Title VARCHAR (50) NOT NULL,
```

```
Mov_Year YEAR NOT NULL,
```

```
Mov_Lang VARCHAR (15) NOT NULL,
```

```
Dir_id INT NOT NULL,
```

```
FOREIGN KEY (Dir_id) REFERENCES DIRECTOR(Dir_id)
```

```
);
```

```
CREATE TABLE MOVIE_CAST (
```

```
Act_id INT,
```

```
Mov_id INT,
```

```
Role VARCHAR (15) NOT NULL,
```

```
PRIMARY KEY (Act_id, Mov_id),
```

```
FOREIGN KEY (Act_id) REFERENCES ACTOR (Act_id),  
FOREIGN KEY (Mov_id) REFERENCES MOVIES(Mov_id)  
);
```

```
CREATE TABLE RATING (  
    Mov_id INT,  
    Rev_starts INT (1) NOT NULL,  
    FOREIGN KEY (Mov_id) REFERENCES (Mov_id)  
);
```

```
INSERT INTO ACTOR VALUES ( 510 , 'Yash' , 'M');  
INSERT INTO DIRECTOR VALUES ( 103 , 'Neel' , '90000001');  
INSERT INTO MOVIES VALUES ( 1008 , 'KGF' , '2018' , 'Kannada' , 103);  
INSERT INTO MOVIE_CAST VALUES ( 510 , 1008 , 'Hero');  
INSERT INTO RATING VALUES ( 1008 , 5);
```

```
SELECT Mov_Title  
FROM MOVIES M, DIRECTOR D  
WHERE M.Dir_id = D.Dir_id AND Dir_Name = 'Mani Rathnam'  
;
```

```
SELECT Mov_Title  
FROM MOVIES M, MOVIE_CAST C  
WHERE M.Mov_id = C.Mov_id AND Act_id IN (  
    SELECT Act_id  
    FROM MOVIE_CAST  
    GROUP BY Act_id  
    HAVING COUNT(Act_id) > 1 )  
;
```

```
SELECT Act_Name
```

```
FROM (ACTOR A JOIN MOVIE_CAST C ON A.Act_id = C.Act_id) JOIN
MOVIES M ON C.Mov_id = M.Mov_id
WHERE M.Mov_Year < 2000 AND Act_Name IN (
SELECT Act_Name
FROM (ACTOR A JOIN MOVIE_CAST C ON A.Act_id = C.Act_id) JOIN
MOVIES M ON C.Mov_id = M.Mov_id WHERE M.Mov_Year > 2024
);
```

```
SELECT Mov_Title, Rev_Stars, MAX(Rev_Stars) FROM MOVIES M,
RATING R
WHERE M.Mov_id = R.Mov_id
GROUP BY Mov_Title
HAVING COUNT(Mov_Title) > 1
ORDER BY Mov_Title;
```

```
UPDATE RATING
SET Rev_Stars = 5
WHERE Mov_id IN (
SELECT Mov_id
FROM MOVIES M, DIRECTOR D
WHERE M.Dir_id = D.Dir_id AND Dir_Name = 'Shankar')
;
```

Program 3 ->

Consider the schema for Employee salary Database:

EMPLOYEE(SSN, Name, Address, Sex, Salary, SuperSSN, DNo)

DEPARTMENT(DNo, DName, MgrSSN, MgrStartDate)

DLOCATION(DNo,DLoc)

PROJECT(PNo, PName, PLocation, DNo)

WORKS_ON(SSN, PNo, Hours)

Write SQL queries to

1. Retrieve the employee numbers of all employees who work on project located in Mysore, Hassan, or Mangalore
2. Retrieve all employees in department 5 whose salary is between 50,000 and 60,000(inclusive)
3. Find the sum of the salaries of all employees, the maximum salary, the minimum salary, and the average salary. Display with all the details of Employee.
4. Select the names of employees whose salary is greater than the average salary of all employees in department 10.
5. For each department having more than 10 employees, retrieve the department no, no of employees drawing more than 40,000 as salary.

```
CREATE TABLE EMPLOYEE (  
    SSN INT PRIMARY KEY,  
    NAME CHAR(10),  
    ADDRESS VARCHAR(15),  
    SEX CHAR(6),  
    SALARY REAL,  
    SUPERSSN INT,  
    DNO INT  
);
```

```
mysql> desc EMPLOYEE;
```

Field	Type	Null	Key	Default	Extra
SSN	int	NO	PRI	NULL	
NAME	char(10)	YES		NULL	
ADDRESS	varchar(15)	YES		NULL	
SEX	char(6)	YES		NULL	
SALARY	double	YES		NULL	
SUPERSSN	int	YES		NULL	
DNO	int	YES		NULL	

```
7 rows in set (0.03 sec)
```

```
CREATE TABLE DEPARTMENT (
    DNO INT PRIMARY KEY,
    DNAME CHAR(15),
    MGRSSN INT,
    MGRSTART DATE
);
```

```
mysql> desc DEPARTMENT;
```

Field	Type	Null	Key	Default	Extra
DNO	int	NO	PRI	NULL	
DNAME	char(15)	YES		NULL	
MGRSSN	int	YES		NULL	
MGRSTART	date	YES		NULL	

```
4 rows in set (0.01 sec)
```

```
CREATE TABLE DLOCATION (
    DNO INT,
    DLOC CHAR(10),
    PRIMARY KEY (DNO, DLOC),
    FOREIGN KEY (DNO) REFERENCES DEPARTMENT(DNO)
);
```

```
mysql> desc DLOCATION;
```

Field	Type	Null	Key	Default	Extra
DNO	int	NO	PRI	NULL	
DLOC	char(10)	NO	PRI	NULL	

```
2 rows in set (0.00 sec)
```

```
CREATE TABLE PROJECT (
    PNO INT PRIMARY KEY,
    PNAME CHAR(15),
    PLOCATION CHAR(10),
    DNO INT,
    FOREIGN KEY (DNO) REFERENCES DEPARTMENT(DNO)
);
```

```
mysql> desc PROJECT;
```

Field	Type	Null	Key	Default	Extra
PNO	int	NO	PRI	NULL	
PNAME	char(15)	YES		NULL	
PLOCATION	char(10)	YES		NULL	
DNO	int	YES	MUL	NULL	

```
4 rows in set (0.00 sec)
```

```
CREATE TABLE WORKS_ON (
    SSN INT,
    PNO INT,
    PRIMARY KEY (SSN, PNO),
    FOREIGN KEY (SSN) REFERENCES EMPLOYEE(SSN),
    FOREIGN KEY (PNO) REFERENCES PROJECT(PNO),
    HOURS INT
);
```

```
mysql> desc WORKS_ON;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| SSN    | int  | NO   | PRI | NULL    |       |
| PNO    | int  | NO   | PRI | NULL    |       |
| HOURS  | int  | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

ALTERING COMMANDS

ALTER TABLE EMPLOYEE ADD CONSTRAINT FOREIGN KEY(DNO)
REFERENCES DEPARTMENT(DNO);

ALTER TABLE DEPARTMENT ADD CONSTRAINT FOREIGN
KEY(MGRSSN) REFERENCES EMPLOYEE(SSN);

INSERT VALUES TO TABLES

INSERT INTO EMPLOYEE (SSN, NAME, ADDRESS, SEX, SALARY)
VALUES

(123, 'John', '123 Elm St', 'Male', 50000),
(987, 'Jane', '456 Oak St', 'Female', 55000),
(112, 'Alice', '789 Pine St', 'Female', 60000),
(556, 'Bob', '101 Maple St', 'Male', 45000),
(667, 'Charlie', '202 Birch St', 'Male', 48000);

```
mysql> select *from EMPLOYEE;
+-----+-----+-----+-----+-----+-----+-----+
| SSN | NAME  | ADDRESS      | SEX  | SALARY | SUPERSSN | DNO |
+-----+-----+-----+-----+-----+-----+-----+
| 112 | Alice | 789 Pine St  | Female | 60000 | NULL     | NULL |
| 123 | John  | 123 Elm St   | Male  | 50000 | NULL     | NULL |
| 556 | Bob   | 101 Maple St | Male  | 45000 | NULL     | NULL |
| 667 | Charlie | 202 Birch St | Male  | 48000 | NULL     | NULL |
| 987 | Jane  | 456 Oak St   | Female | 55000 | NULL     | NULL |
+-----+-----+-----+-----+-----+-----+-----+
5 rows in set (0.01 sec)
```

INSERT INTO DEPARTMENT (DNO, DNAME, MGRSSN, MGRSTART)
VALUES

(1, 'HR', 123, '2015-06-01'),
(2, 'Finance', 987, '2017-04-15'),

(3, 'Engineering', 112, '2018-08-23'),
(4, 'Sales', 556, '2016-02-10'),
(5, 'Marketing', 667, '2020-11-30');

mysql> select *from DEPARTMENT;

DNO	DNAME	MGRSSN	MGRSTART
1	HR	123	2015-06-01
2	Finance	987	2017-04-15
3	Engineering	112	2018-08-23
4	Sales	556	2016-02-10
5	Marketing	667	2020-11-30

5 rows in set (0.01 sec)

UPDATING “DNO”

UPDATE EMPLOYEE SET
DNO='3'
WHERE SSN='123';

UPDATE EMPLOYEE SET
DNO='1'
WHERE SSN='987';

UPDATE EMPLOYEE SET
DNO='2'
WHERE SSN='112';

UPDATE EMPLOYEE SET
DNO='5'

WHERE SSN='556';

UPDATE EMPLOYEE SET

DNO='5'

WHERE SSN='667';

```
mysql> select *from EMPLOYEE;
+-----+-----+-----+-----+-----+-----+-----+
| SSN | NAME | ADDRESS | SEX | SALARY | SUPERSSN | DNO |
+-----+-----+-----+-----+-----+-----+-----+
| 112 | Alice | 789 Pine St | Female | 60000 | NULL | 2 |
| 123 | John | 123 Elm St | Male | 50000 | NULL | 3 |
| 556 | Bob | 101 Maple St | Male | 45000 | NULL | 5 |
| 667 | Charlie | 202 Birch St | Male | 48000 | NULL | 5 |
| 987 | Jane | 456 Oak St | Female | 55000 | NULL | 1 |
+-----+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

INSERT INTO DLOCATION (DNO, DLOC) VALUES

(1, 'New York'),

(2, 'Chicago'),

(3, 'San Franc'),

(4, 'Los Angel'),

(5, 'Miami');

```
mysql> select *from DLOCATION;
+-----+-----+
| DNO | DLOC |
+-----+-----+
| 1 | New York |
| 2 | Chicago |
| 3 | San Franc |
| 4 | Los Angel |
| 5 | Miami |
+-----+-----+
5 rows in set (0.00 sec)
```

INSERT INTO PROJECT (PNO, PNAME, PLOCATION, DNO) VALUES

(1, 'Project Alpha', 'Mysuru', 1),
 (2, 'Project Beta', 'Hassan', 2),
 (3, 'Project Gamma', 'Mangalore', 3),
 (4, 'Project Alpha', 'Mysuru', 4),
 (5, 'Project Beta', 'Hassan', 5);

```
mysql> select *from PROJECT;
```

PNO	PNAME	PLOCATION	DNO
1	Project Alpha	Mysuru	1
2	Project Beta	Hassan	2
3	Project Gamma	Mangalore	3
4	Project Alpha	Mysuru	4
5	Project Beta	Hassan	5

5 rows in set (0.00 sec)

INSERT INTO WORKS_ON (SSN, PNO, HOURS) VALUES

(123, 1, 40),
 (987, 2, 35),
 (112, 3, 30),
 (556, 1, 25),
 (667, 2, 40);

```
mysql> select *from WORKS_ON
```

SSN	PNO	HOURS
112	3	30
123	1	40
556	1	25
667	2	40
987	2	35

5 rows in set (0.00 sec)

1. Retrieve the employee numbers of all employees who work on project located in Mysore, Hassan, or

Mangalore

```
SELECT COUNT(*)
FROM EMPLOYEE E, PROJECT P, WORKS_ON W
WHERE E.SSN = W.SSN
AND W.PNO = P.PNO
AND (P. PLOCATION = 'MYSORE' OR P. PLOCATION = 'BENGALURU'
OR P. PLOCATION = 'MANGALORE');
```

```
+-----+
| COUNT(*) |
+-----+
|          1 |
+-----+
1 row in set (0.01 sec)
```

2. Retrieve all employees in department 5 whose salary is between 50,000 and 60,000(inclusive)

select *

from EMPLOYEE

WHERE(SALARY BETWEEN 40000 AND 60000)AND DNO=5;

```
+-----+-----+-----+-----+-----+-----+-----+
| SSN | NAME   | ADDRESS      | SEX  | SALARY | SUPERSSN | DNO |
+-----+-----+-----+-----+-----+-----+-----+
| 556 | Bob    | 101 Maple St | Male | 45000  | NULL     | 5   |
| 667 | Charlie| 202 Birch St | Male | 48000  | NULL     | 5   |
+-----+-----+-----+-----+-----+-----+-----+
2 rows in set (0.01 sec)
```

3. Find the sum of the salaries of all employees, the maximum salary, the minimum salary, and the average salary. Display with all the details of Employee.

```
SELECT SUM(E.SALARY), MAX(E.SALARY), MIN(E.SALARY),
AVG(E.SALARY)
```

FROM EMPLOYEE E, DEPARTMENT D

SUM(E.SALARY)	MAX(E.SALARY)	MIN(E.SALARY)	AVG(E.SALARY)
1290000	60000	45000	51600

1 row in set (0.01 sec)

4. Select the names of employees whose salary is greater than the average salary of all employees in department 10.

```
SELECT NAME
FROM EMPLOYEE E
WHERE SALARY > (SELECT AVG(SALARY)
                FROM EMPLOYEE
                WHERE DNO = 5);
```

NAME
Alice
John
Charlie
Jane

4 rows in set (0.01 sec)

5. For each department having more than 10 employees, retrieve the department no, no of employees drawing more than 40,000 as salary.

```
SELECT DNO, COUNT(*) AS num_employees_above_40000
FROM EMPLOYEE
WHERE salary > 40000
GROUP BY DNO
HAVING COUNT(SSN) > 2;
```

```
+-----+-----+
| DNO | num_employees_above_40000 |
+-----+-----+
|    5 |                          2 |
+-----+-----+
1 row in set (0.00 sec)
```

EXPERIMENT – 4

Consider the schema for Bus ticket reservation Database:

PASSENGER (P_id, P_Name, P_Gender,P_city)

AGENCY(A_id,A_Name,A_city)

BUS(B_id,B_date, B_time, B_scr, B_dest) BOOKING(P_id,
A_id,B_id,B_date,B_time) Write SQL queries to

1. Get the Complete Details of all the Buses to MIT-Mysore.
2. Find only the Bus Number for Passenger with PID 123 for buses to Mandya, before 05/05/2024.
3. Find the Passenger Name for those who don't have any booking in any buses.
4. Get the Details of the buses that are scheduled on both dates 01/02/2024 and 02/02/2024 at 16:00 hrs.
5. Find the details of male passengers who are associated with “Happy Smiling” Agency

```
CREATE TABLE PASSENGER ( P_id INT PRIMARY KEY, P_Name  
VARCHAR(100), P_Gender VARCHAR(10), P_city VARCHAR(100)  
);
```

```
CREATE TABLE AGENCY ( A_id INT PRIMARY KEY, A_Name  
VARCHAR(100), A_city VARCHAR(100)  
);
```

```
CREATE TABLE BUS (  
  B_id INT, B_date DATE, B_time TIME,  
  B_scr VARCHAR(100), B_dest VARCHAR(100),  
  PRIMARY KEY (B_id, B_date, B_time)  
);
```

```
CREATE TABLE BOOKING (  
  P_id INT, A_id INT, B_id INT, B_date DATE, B_time TIME,  
  PRIMARY KEY (P_id, A_id, B_id, B_date, B_time), FOREIGN KEY (P_id)  
  REFERENCES PASSENGER(P_id), FOREIGN KEY (A_id) REFERENCES  
  AGENCY(A_id),  
  FOREIGN KEY (B_id, B_date, B_time) REFERENCES BUS(B_id, B_date,  
  B_time)  
);
```

```
mysql> desc PASSENGER;
```

Field	Type	Null	Key	Default	Extra
P_id	int	NO	PRI	NULL	
P_Name	varchar(100)	YES		NULL	
P_Gender	varchar(10)	YES		NULL	
P_city	varchar(100)	YES		NULL	

```
4 rows in set (0.04 sec)
```

```
mysql> DESC AGENCY;
```

Field	Type	Null	Key	Default	Extra
A_id	int	NO	PRI	NULL	
A_Name	varchar(100)	YES		NULL	
A_city	varchar(100)	YES		NULL	

```
3 rows in set (0.00 sec)
```

```
mysql> DESC BUS;
```

Field	Type	Null	Key	Default	Extra
B_id	int	NO	PRI	NULL	
B_date	date	NO	PRI	NULL	
B_time	time	NO	PRI	NULL	
B_scr	varchar(100)	YES		NULL	
B_dest	varchar(100)	YES		NULL	

```
5 rows in set (0.00 sec)
```

```
mysql> DESC BOOKING;
```

Field	Type	Null	Key	Default	Extra
P_id	int	NO	PRI	NULL	
A_id	int	NO	PRI	NULL	
B_id	int	NO	PRI	NULL	
B_date	date	NO	PRI	NULL	
B_time	time	NO	PRI	NULL	

```
5 rows in set (0.00 sec)
```

```
INSERT INTO PASSENGER VALUES (101, 'Alice', 'Female', 'Bangalore');
INSERT INTO PASSENGER VALUES (102, 'Bob', 'Male', 'Mysore');

INSERT INTO PASSENGER VALUES (103, 'Charlie', 'Male', 'Mandya');
INSERT INTO PASSENGER VALUES (104, 'Diana', 'Female', 'Tumkur');
INSERT INTO PASSENGER VALUES (123, 'Ethan', 'Male', 'Mandya');
```

```
mysql> SELECT *FROM PASSENGER;
+-----+-----+-----+-----+
| P_id | P_Name | P_Gender | P_city |
+-----+-----+-----+-----+
| 101  | Alice  | Female   | Bangalore |
| 102  | Bob    | Male     | Mysore    |
| 103  | Charlie | Male     | Mandya    |
| 104  | Diana  | Female   | Tumkur    |
| 123  | Ethan  | Male     | Mandya    |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

```
INSERT INTO AGENCY VALUES (1, 'Speed Travels', 'Bangalore'); INSERT
INTO AGENCY VALUES (2, 'Happy Smiling', 'Mysore');

INSERT INTO AGENCY VALUES (3, 'Green Wheels', 'Mandya');
```

```
mysql> SELECT *FROM AGENCY;
+-----+-----+-----+
| A_id | A_Name       | A_city |
+-----+-----+-----+
| 1    | Speed Travels | Bangalore |
| 2    | Happy Smiling | Mysore    |
| 3    | Green Wheels  | Mandya    |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

-- Buses to MIT-Mysore

```
INSERT INTO BUS VALUES (201, '2024-04-15', '10:00:00', 'Bangalore', 'MIT-Mysore');
```

```
INSERT INTO BUS VALUES (202, '2024-04-20', '11:30:00', 'Mandya', 'MIT-Mysore');
```

-- Buses to Mandya

```
INSERT INTO BUS VALUES (203, '2024-04-30', '09:00:00', 'Mysore', 'Mandya');
```

```
INSERT INTO BUS VALUES (204, '2024-05-06', '14:00:00', 'Tumkur', 'Mandya');
```

-- For query 4: buses scheduled at 16:00 on both dates

```
INSERT INTO BUS VALUES (205, '2024-02-01', '16:00:00', 'Mysore', 'Bangalore');
```

```
INSERT INTO BUS VALUES (205, '2024-02-02', '16:00:00', 'Mysore', 'Bangalore');
```

```
mysql> SELECT *FROM BUS;
```

B_id	B_date	B_time	B_scr	B_dest
201	2024-04-15	10:00:00	Bangalore	MIT-Mysore
202	2024-04-20	11:30:00	Mandya	MIT-Mysore
203	2024-04-30	09:00:00	Mysore	Mandya
204	2024-05-06	14:00:00	Tumkur	Mandya
205	2024-02-01	16:00:00	Mysore	Bangalore
205	2024-02-02	16:00:00	Mysore	Bangalore

```
6 rows in set (0.00 sec)
```

-- Bookings by Ethan (P_id = 123)

```
INSERT INTO BOOKING VALUES (123, 2, 203, '2024-04-30', '09:00:00');
```

-- Bookings by Bob via Happy Smiling

```
INSERT INTO BOOKING VALUES (102, 2, 201, '2024-04-15', '10:00:00');
```

-- Bookings by Charlie via Green Wheels

```
INSERT INTO BOOKING VALUES (103, 3, 202, '2024-04-20', '11:30:00');
```

```
mysql> SELECT *FROM BOOKING;
+-----+-----+-----+-----+-----+
| P_id | A_id | B_id | B_date      | B_time      |
+-----+-----+-----+-----+-----+
| 102  | 2    | 201  | 2024-04-15  | 10:00:00    |
| 123  | 2    | 203  | 2024-04-30  | 09:00:00    |
| 103  | 3    | 202  | 2024-04-20  | 11:30:00    |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

QUERIES:

1. Get the Complete Details of all the Buses to MIT-Mysore.

```
SELECT *
FROM BUS
WHERE B_dest = 'MIT-Mysore';
```

```
+-----+-----+-----+-----+-----+
| B_id | B_date      | B_time      | B_scr      | B_dest      |
+-----+-----+-----+-----+-----+
| 201  | 2024-04-15  | 10:00:00    | Bangalore  | MIT-Mysore  |
| 202  | 2024-04-20  | 11:30:00    | Mandya     | MIT-Mysore  |
+-----+-----+-----+-----+-----+
2 rows in set (0.03 sec)
```

2. Find only the Bus Number for Passenger with PID 123 for buses to Mandya, before 05/05/2024

```
SELECT B.B_id FROM BUS B
JOIN BOOKING BK ON B.B_id = BK.B_id AND B.B_date = BK.B_date AND
B.B_time = BK.B_time WHERE BK.P_id = 123
AND B.B_dest = 'Mandya'
AND B.B_date < '2024-05-05';
```

```

-> AND B.B_date <
+-----+
| B_id |
+-----+
| 203 |
+-----+
1 row in set (0.00 sec)

```

3. Find the Passenger Name for those who don't have any booking in any buses

```

SELECT P.P_Name
FROM PASSENGER P WHERE P.P_id NOT IN (
SELECT P_id FROM BOOKING
);

```

```

-> );
+-----+
| P_Name |
+-----+
| Alice |
| Diana |
+-----+
2 rows in set (0.01 sec)

```

4. Get the Details of the buses that are scheduled on both dates 01/02/2024 and 02/02/2024 at 16:00 hrs

```

SELECT *
FROM BUS
WHERE B_time = '16:00:00' AND B_id IN (
SELECT B_id FROM BUS
WHERE B_date = '2024-02-01' INTERSECT

```

```

SELECT B_id FROM BUS
WHERE B_date = '2024-02-02'
);

```

```

+-----+-----+-----+-----+-----+
| B_id | B_date   | B_time   | B_scr | B_dest   |
+-----+-----+-----+-----+-----+
| 205  | 2024-02-01 | 16:00:00 | Mysore | Bangalore |
| 205  | 2024-02-02 | 16:00:00 | Mysore | Bangalore |
+-----+-----+-----+-----+-----+
2 rows in set (0.02 sec)

```

5. Find the details of male passengers who are associated with “Happy Smiling” Agency

```

SELECT DISTINCT P.* FROM PASSENGER P
JOIN BOOKING BK ON P.P_id = BK.P_id JOIN AGENCY A ON BK.A_id =
A.A_id
WHERE P.P_Gender = 'Male'
AND A.A_Name = 'Happy Smiling';

```

```

+-----+-----+-----+-----+
| P_id | P_Name | P_Gender | P_city |
+-----+-----+-----+-----+
| 102  | Bob    | Male     | Mysore |
| 123  | Ethan  | Male     | Mandya |
+-----+-----+-----+-----+
2 rows in set (0.03 sec)

```

EXPERIMENT -5

Consider the schema for Matrimonial Database :

ENROLL(E_Name,E_Gender,E_Age,E_Qualification, E_Salary,E_Address, E_City) WORKS(E_name, E_salary, E_city)

MARRIAGE_BUREAU(MB_name,MB_city,MB_charge) Write SQL queries to :

1. Find the Names and Cities for all the Groom, who work for MIT-Mysore and earn more than Rs.60,000/- as salary.
2. Find the Company that has the Least Fee for Marriage Service.
3. Find the name of all the Brides in the database who live in the same cities and on the same street
as do their Groom.
4. Find the names of Groom in the database, whose qualification and age is same as bride.
5. Find the name of the Groom in the database, who earns more than all Bride lives in "Mandya"

```
-- Table 1: ENROLL CREATE TABLE ENROLL (  
E_Name VARCHAR(100) PRIMARY KEY,  
E_Gender CHAR(1) CHECK (E_Gender IN ('M', 'F')), E_Age INT CHECK  
(E_Age >= 18),  
E_Qualification VARCHAR(100), E_Salary DECIMAL(10, 2),  
E_Address VARCHAR(255), E_City VARCHAR(100)  
);
```

```
-- Table 2: WORKS CREATE TABLE WORKS (  
E_Name VARCHAR(100), E_Salary DECIMAL(10, 2), E_City  
VARCHAR(100),
```

```
FOREIGN KEY (E_Name) REFERENCES ENROLL(E_Name)
);
```

```
-- Table 3: MARRIAGE_BUREAU CREATE TABLE MARRIAGE_BUREAU
(
MB_Name VARCHAR(100) PRIMARY KEY, MB_City VARCHAR(100),
MB_Charge DECIMAL(10, 2)
);
```

```
mysql> desc works;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| E_Name | varchar(100)  | YES  | MUL | NULL    |       |
| E_Salary | decimal(10,2) | YES  |     | NULL    |       |
| E_City  | varchar(100)  | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> desc enroll;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| E_Name | varchar(100)  | NO   | PRI | NULL    |       |
| E_Gender | char(1)       | YES  |     | NULL    |       |
| E_Age  | int           | YES  |     | NULL    |       |
| E_Qualification | varchar(100) | YES  |     | NULL    |       |
| E_Salary | decimal(10,2) | YES  |     | NULL    |       |
| E_Address | varchar(255)  | YES  |     | NULL    |       |
| E_City  | varchar(100)  | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.07 sec)
```

```
mysql> desc marriage_bureau;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| MB_Name | varchar(100)  | NO   | PRI | NULL    |       |
| MB_City | varchar(100)  | YES  |     | NULL    |       |
| MB_Charge | decimal(10,2) | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

INSERT INTO ENROLL VALUES

('Arun', 'M', 30, 'B.Tech', 75000, '5th Cross, MG Road', 'Mysore'),
('Ravi', 'M', 28, 'MBA', 65000, '1st Main, VV Mohalla', 'MIT-Mysore'),
('Raj', 'M', 32, 'M.Tech', 90000, 'Jayanagar, 3rd Block', 'Mandya'),
('Priya', 'F', 30, 'B.Tech', 50000, '5th Cross, MG Road', 'Mysore'),
('Sneha', 'F', 28, 'MBA', 55000, '1st Main, VV Mohalla', 'MIT-Mysore'),
('Anita', 'F', 32, 'M.Tech', 60000, 'Jayanagar, 3rd Block', 'Mandya');

INSERT INTO WORKS VALUES

('Arun', 75000, 'Infosys-Mysore'),
('Ravi', 65000, 'MIT-Mysore'),
('Raj', 90000, 'IBM-Mandya'),
('Priya', 50000, 'Wipro-Mysore'),
('Sneha', 55000, 'MIT-Mysore'),
('Anita', 60000, 'IBM-Mandya');

INSERT INTO MARRIAGE_BUREAU VALUES

('Shaadi Center', 'Bangalore', 20000), ('MatchMakers', 'Mysore', 15000),
('Elite Matrimony', 'Mandya', 18000), ('Happy Couples', 'Mysore', 14000);

```
mysql> select * from enroll;
```

E_Name	E_Gender	E_Age	E_Qualification	E_Salary	E_Address	E_City
Anita	F	32	M.Tech	60000.00	Jayanagar, 3rd Block	Mandya
Arun	M	30	B.Tech	75000.00	5th Cross, MG Road	Mysore
Priya	F	30	B.Tech	50000.00	5th Cross, MG Road	Mysore
Raj	M	32	M.Tech	90000.00	Jayanagar, 3rd Block	Mandya
Ravi	M	28	MBA	65000.00	1st Main, VV Mohalla	MIT-Mysore
Sneha	F	28	MBA	55000.00	1st Main, VV Mohalla	MIT-Mysore

```
6 rows in set (0.00 sec)
```

```
mysql> select * from works;
```

E_Name	E_Salary	E_City
Arun	75000.00	Infosys-Mysore
Ravi	65000.00	MIT-Mysore
Raj	90000.00	IBM-Mandya
Priya	50000.00	Wipro-Mysore
Sneha	55000.00	MIT-Mysore
Anita	60000.00	IBM-Mandya

```
6 rows in set (0.00 sec)
```

```
mysql> select * from marriage_bureau;
```

MB_Name	MB_City	MB_Charge
Elite Matrimony	Mandya	18000.00
Happy Couples	Mysore	14000.00
MatchMakers	Mysore	15000.00
Shaadi Center	Bangalore	20000.00

```
4 rows in set (0.00 sec)
```

1. Find the Names and Cities for all the Groom, who work for MIT-Mysore and earn more than Rs.60,000/- as salary.

```
SELECT E.E_Name, E.E_City FROM ENROLL E
JOIN WORKS W ON E.E_Name = W.E_Name
WHERE E.E_Gender = 'M'
AND W.E_City = 'MIT-Mysore' AND W.E_Salary > 60000;
```

```
+-----+-----+
| E_Name | E_City      |
+-----+-----+
| Ravi   | MIT-Mysore  |
+-----+-----+
1 row in set (0.00 sec)
```

1. Find the Company that has the Least Fee for Marriage Service.

```
SELECT MB_Name
FROM MARRIAGE_BUREAU
WHERE MB_Charge = ( SELECT MIN(MB_Charge)
FROM MARRIAGE_BUREAU
);
```

```
+-----+
| MB_Name      |
+-----+
| Happy Couples |
+-----+
1 row in set (0.01 sec)
```

2. Find the name of all the Brides in the database who live in the same cities and on the same street as do their Groom.

3.

```
SELECT DISTINCT B.E_Name AS Bride_Name FROM ENROLL B  
JOIN ENROLL G ON B.E_City = G.E_City AND B.E_Address = G.E_Address  
WHERE B.E_Gender = 'F' AND G.E_Gender = 'M';
```

```
+-----+  
| Bride_Name |  
+-----+  
| Priya      |  
| Anita      |  
| Sneha      |  
+-----+  
3 rows in set (0.00 sec)
```

4. Find the names of Groom in the database, whose qualification and age is same as bride.

```
SELECT DISTINCT G.E_Name AS Groom_Name FROM ENROLL G  
JOIN ENROLL B ON G.E_Qualification = B.E_Qualification AND G.E_Age =  
B.E_Age WHERE G.E_Gender = 'M' AND B.E_Gender = 'F';
```

```
-> WHERE G.E_Gender = 'M' -  
+-----+  
| Groom_Name |  
+-----+  
| Raj        |  
| Arun       |  
| Ravi       |  
+-----+  
3 rows in set (0.00 sec)
```

5. Find the name of the Groom in the database, who earns more than all Brides living in “Mandya”

```
SELECT E_Name FROM ENROLL
WHERE E_Gender = 'M' AND E_Salary > ALL ( SELECT E_Salary
FROM ENROLL
WHERE E_Gender = 'F' AND E_City = 'Mandya'
);
```

```
| E_Name |
+-----+
| Arun   |
| Raj    |
| Ravi   |
+-----+
3 rows in set (0.00 sec)
```